

QGIS Contour Map Cheat Sheet

Padstow Estuary workflow for cleaning contours for slider overlays

Goal: keep the useful 0.5 m dune / sandbank contours, remove the sea rubbish, tidy tiny fragments, and export a clean transparent PNG for the web app.

1. Duplicate the contour layer

- Right-click the contour layer in Layers > Duplicate Layer.
- Rename the copy something like Contours_cleaned or Contours_slider_test.
- Always work on the copy so the original stays untouched.

2. Remove the sea / bad low contours

- Right-click the copied contour layer > Filter.
- Use the contour height field (often ELEV) and keep only practical values.
- Example expressions: "ELEV" >= -2 or "ELEV" >= 0 or "ELEV" >= 2
- Use the one that removes the noisy water contours but keeps the useful banks.

3. Remove tiny contour fragments

- Open the copied contour layer attribute table.
- Use Select by Expression and test the tiny line fragments with:
- `$length < 5`
- If the selection only grabs the small rubbish, turn editing on and press Delete.
- Save edits. Leave the threshold at 5 if anything higher starts removing important main contours.

4. Deselect everything

- Press Ctrl + Shift + A to clear the selection.

5. Optional smoothing test

- Duplicate the cleaned contour layer again before smoothing.
- Open Processing > Toolbox > search for Smooth.
- Run one light pass first (Iterations: 1).
- Compare against the unsmoothed version. Keep it only if it softens the twitchy wiggles without damaging the main contour shape.

6. Optional simplify test

- Only try this after smoothing, and only very lightly.
- Processing > Toolbox > search for Simplify.
- Use a small tolerance and compare carefully. Stop if the lines start looking angular or lose the bank shape.

7. Label check

- If labels show huge 6-digit numbers, the layer is labelling the wrong field.
- Contour labels should come from the elevation field, not fid / id / objectid.

- For web overlays, keep labels off or only label major contours to avoid clutter.

8. Export for the slider / overlay

- Style the final contour layer cleanly.
- Export with a transparent background if you want the lines to sit over the ortho or height map.
- Keep the output simple and client-friendly: fewer labels, cleaner lines, no water rubbish.

Quick reference

Task	What to use
Filter out bad sea contours	"ELEV" >= -2 / "ELEV" >= 0 / "ELEV" >= 2
Select tiny fragments	\$length < 5
Deselect all	Ctrl + Shift + A
Delete selected features	Toggle Editing on > Delete > Save edits
Smoothing	Processing Toolbox > Smooth > Iterations: 1
Simplify	Processing Toolbox > Simplify > very light test only

Best current working rule: keep the threshold at 5 for tiny fragments, because anything above that starts removing the main contours that matter.